

# Moosa Abbasi

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## Education

### University of South Florida, Honors College

Tampa, FL

Bachelor of Science in Computer Science

Aug. 2023 – Aug. 2027

GPA: 3.90/4.00 | Green and Gold Presidential Scholar, Dean's List

**Relevant Coursework:** Data Structures, Analysis of Algorithms, Computer Architecture, Software Engineering, Computer Logic & Design, Automata Theory, Introduction to Artificial Intelligence, Discrete Structures

## Technical Skills

**Languages & Databases:** Python, C, C++, JavaScript, SQL, DynamoDB, MySQL

**Frameworks & Libraries:** Streamlit, React.js, OpenCV, Pandas, NumPy, Plotly, scikit-learn, XGBoost, Prophet

**Cloud & DevOps:** AWS (Lambda, API Gateway, S3, CloudWatch, IAM), Docker, Git, GitHub Actions

**Security & Testing:** Burp Suite, OWASP ZAP, OWASP WSTG v4.2, Playwright, HTTP Traffic Analysis, XSS/SQLi, Prompt Injection

**Certifications:** Machine Learning Specialization (Supervised ML, Advanced Learning Algorithms, Unsupervised Learning) — Andrew Ng, DeepLearning.AI & Stanford University, Coursera · 2026

## Experience

### Undergraduate Research Assistant — AI Course Companion (AICC)

May 2026 – Present

University of South Florida | CIS 4915 | Dr. Oguzhan Topsakal

- Developed **7 Playwright** automated scripts executing **98 black-box test cases** across **5 OWASP WSTG v4.2** phases; discovered **5 XSS payloads** crashing the backend with raw **SERVER\_ERROR**, **6 missing HTTP security headers (CSP, HSTS, X-Frame-Options)**, and **no CSRF protection** across **7 SameSite=None** cookies lacking **HttpOnly** with **390-day** lifetimes.
- Gray-box source review uncovered **7 unauthenticated API endpoints** with no **validateSession()** checks, reproduced live unauthenticated course content dump and spoofed instructor write; identified **deterministic HMAC-SHA256 session IDs** keyed to Canvas user ID, enabling session forgery if the secret leaks.

### Project Management Intern

May 2024 – Aug 2024

Huawei Technologies

- Resolved visibility gaps across **5+ concurrent DMD projects** by building structured **tracking dashboards** that standardized reporting cycles and cross-team workflows, improving on-time delivery by **20%**.
- Eliminated siloed communication by introducing a **data-driven reporting** system and standardized **project documentation** workflow, reducing delays by **25%** and boosting team efficiency by **15%**.

## Projects

### Smart Job Tracker

May 2025 – Present

AWS Lambda, DynamoDB, API Gateway, Cognito, SES, Chrome Extension (MV3), Streamlit, Python

Tampa, FL

- Eliminated manual logging via a **Chrome Extension (MV3)** with a **3-layer detection system** using **MAIN world XHR monkey-patching** and **MutationObserver**, plus a **chrome.alarms** queue guaranteeing **zero application loss** on service worker suspension.
- Engineered a **5-Lambda serverless backend** with **DynamoDB user-scoped isolation**, **API Gateway REST API** protected by a **Cognito JWT** authorizer, **preemptive token refresh** with race-condition locking, and automatic **401 retry** across all 3 clients.
- Deployed a live **Streamlit** dashboard with **EventBridge + SES** daily email reminders for application tracking; reduced total operating cost to **\$0/month** by architecting entirely within the **AWS Free Tier**.

### USF Emissions Analysis Dashboard

Jan 2026 – Present

Python, Streamlit, Plotly, Pandas, scikit-learn, XGBoost, Prophet, GHG Protocol

Tampa, FL

- Quantified **~83.8M kg CO<sub>2</sub>** across **5 categories** (Scope 1 + Scope 3 Cat 4/6/7/9) by building a full-stack **GHG emissions dashboard** in **Python/Streamlit**, applying **GHG Protocol** and **EPA SmartWay** emission factors to **~50,000 synthetic records**.
- Built **3 ML models: IsolationForest** anomaly detection flagging **500 shipments** with **6.3M kg CO<sub>2</sub>** savings potential, **Prophet** 24-month emissions forecasting, and **XGBoost** regression at **R<sup>2</sup>=0.9965** with invoice amount as the top emission predictor.
- Deployed on **Streamlit Cloud** via **GitHub** with automated redeployment; presented findings to an airport industry client in collaboration with **Patel's College of Global Sustainability** for Scope 1 and Scope 3 strategy recommendations.

### Digital Image Processing (DIP) System

Jan 2025 – Apr 2025

Python, OpenCV, NumPy, Matplotlib, Image Segmentation

Tampa, FL

- Improved edge detection accuracy by **35%** and visual precision by **40%** across **500+ datasets** by engineering a preprocessing pipeline using **Python, OpenCV, Gaussian/Sobel** filtering, and adaptive **Otsu thresholding** algorithms.
- Reduced processing latency by **30%** by building a modular real-time analysis pipeline with **Matplotlib** visualization and automated benchmarking, enabling efficient model evaluation and performance debugging.
- Evaluated **5+ segmentation techniques**, including **K-means clustering**, **Watershed**, and **morphological operations**, against ground truth masks, quantifying boundary detection quality across varied noise and lighting conditions.